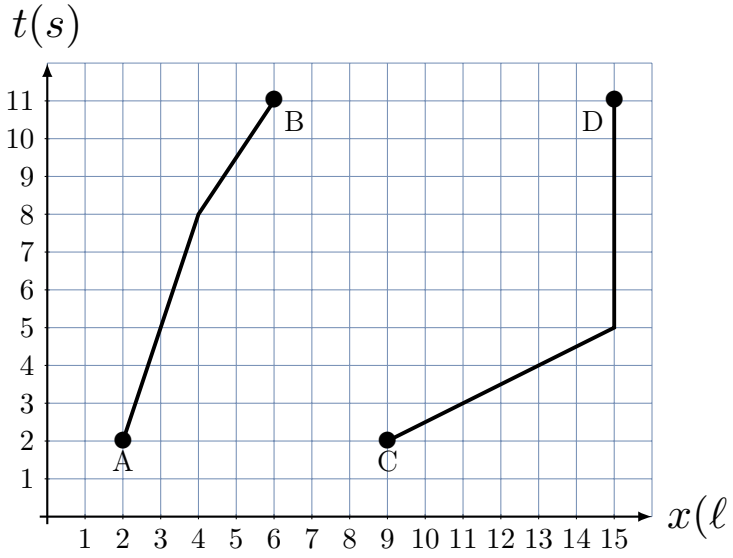
1. Twin Alice stays home, Twin Bob travels to Alpha Centauri, which is 4.37 Ly away on a rocket at 0.96c. Bob stays at Alpha Centauri for 5 years collecting data and travels home on a rocket at 0.82c. If bob departed on the twin's 30th birthday, how old is each when he returns?

2. A starship voyages to a distant planet 10 ly away. The explorers stay 1 year, return at the same speed, and arrive back on earth 26 years, as measured on earth, after they left. Assume that the time needed to accelerate and decelerate is negligible. (a) What is the speed of the starship, and (b) How much time has elapsed on the astronauts choronometers?

3. Two worldlines are shown. (a) How much time elapses for the person traveling between A and B? (b) What is the spacetime interval between A and B? (c) Why is the path between C and D impossible?



4. An frame S event A happens at (t,x) =(5,10). A rocket (Frame S')zooms by going $\beta = 5/12$ in the +x direction. The two frames coincide at t=0.

A. Use a Lorentz transformation to find the spacetime coordinates of event A in S'

B. Use a piece of tilted graph paper to construct and calibrate your spacetime diagram. Show that the spacetime coordinate is approximately the same as your result for part **A**.

5. Discuss the saying: The future is not there awaiting us, but is something we make as we go along.